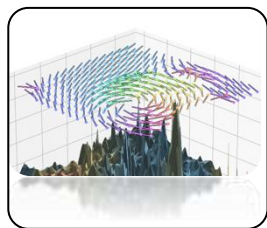
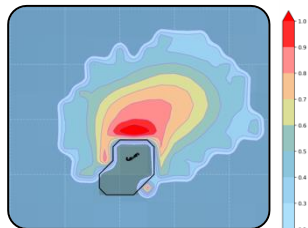


Data-Grounded Ocean Environment



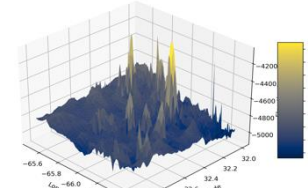
Currents & Tides



Pollution & Elements



Marine Organisms



Bathymetry & Terrain

Unified Ocean Environment
From (Online Demo)

shared grid, aligned meta data, reproducible variants

Environment Query Interface & Tasks Design

Navigation / Tracking

Sensing / Inspection

Coordination / Pollution

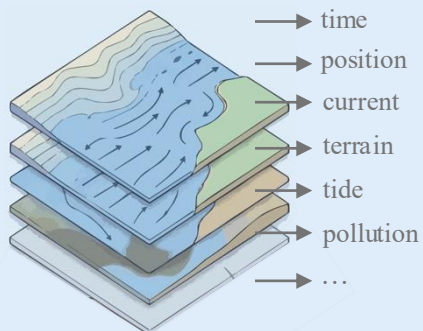
- *same tasks*
- *same tiered difficulty*
- *same grounding logic*
- *comparable across backends*

Tier 1

Tier 2

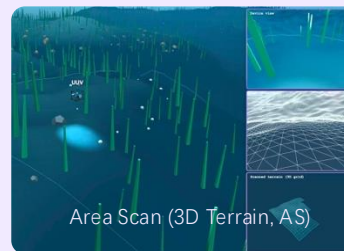
Tier 3

Ocean Signal= Q (position, time, depth, context)



Observations O_t

Canonical Task Ladder + Agent Interaction



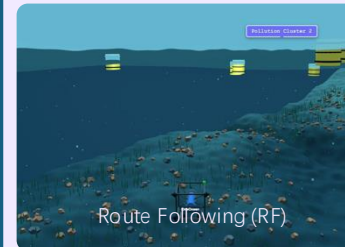
Area Scan (3D Terrain, AS)



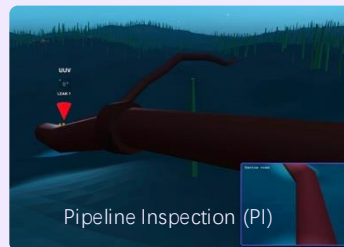
Underwater Pollution Lift (UPL)



Fish Protection / Herding (FH)



Route Following (RF)



Pipeline Inspection (PI)



Surface Pollution Cleanup (SPC)

Policy / Controller

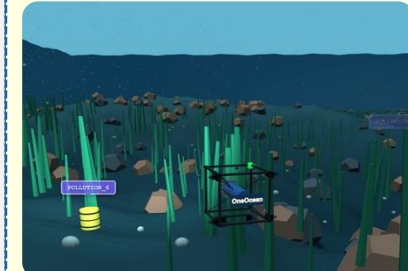
- ❖ Heuristic
 - ❖ Learned, LLM high-level planner + low-level controller
- Optional Memory / Planning State*

Action a_t

- ❖ bounded control command
- Same task contract across both backends*

Agent Interaction

Core Benchmark Backend



3–6 DoF dynamics | 10+ Tasks,
Single-agent / Multi-agent (2-10)

External Scene Backend

OceanGym + HoloOcean 2.0



Standardized Outputs

- Standardized Metrics
- Success Rate / Tracking Error
- Episode Time / Energy Usage
- Collisions / Violations
- Coordination Cost